

Client Case Study: Atlas Manufacturing Co.

Manufacturing | Cedar Rapids, IA

The Challenge

Atlas Manufacturing Co. (client ID CL-1004), a mid-sized producer of precision hydraulic components, had relied on paper travelers and manual inspection logs across its three production lines in Cedar Rapids since the plant opened in the 1980s. Leadership estimated that unplanned downtime was costing the facility approximately \$4.2 million annually, but lacked the instrumented data to confirm root causes or prioritize interventions. Atlas engaged CRCG in February 2022 to design and implement a plant floor digitization program (Project PRJ-2201) under the Operations Excellence practice's OE-210 offering.

Our Approach

Phase 1: Current-State Assessment (Weeks 1–6)

CRCG's team, led by Senior Engagement Partner Priya Natarajan, conducted a six-week diagnostic combining time-and-motion studies, machine downtime log review (where logs existed), and operator interviews across all three lines. The assessment identified that Line 2, the oldest of the three lines, accounted for nearly 58% of total unplanned downtime despite producing only 31% of unit volume.

Phase 2: Sensor and MES Deployment (Weeks 7–28)

The team designed a phased IoT sensor rollout, prioritizing Line 2, paired with selection and deployment of a Manufacturing Execution System (MES) to replace paper travelers. CRCG ran a structured vendor evaluation across four MES candidates before Atlas selected a mid-market MES platform; CRCG's team configured the work-order and quality-inspection modules and integrated sensor telemetry into a unified downtime-cause dashboard.

Phase 3: Hypercare and Capability Transfer (Weeks 29–40)

Following pilot go-live on Line 2 in August 2022, CRCG provided eight weeks of hypercare support alongside a structured training curriculum for shift supervisors, before extending the rollout to Lines 1 and 3. The engagement formally closed in November 2022, with a capability hand-off plan transitioning dashboard ownership to Atlas's newly created Continuous Improvement Manager role.

Results

Metric	Before	After	Notes
Unplanned downtime (Line 2)	31%	14%	Measured over trailing 90 days post-rollout
Plant-wide unplanned downtime cost	\$4.2M/yr (est.)	\$2.6M/yr (est.)	Estimate per Atlas finance team, not independently audited by CRCG

Metric	Before	After	Notes
Work order cycle time	3.4 days	1.1 days	Paper traveler elimination
Quality escapes (defects reaching customer)	0.81%	0.46%	Trailing 6-month average

“CRCG didn’t just hand us a dashboard and leave. They sat on the floor with our supervisors for two months until the new way of working actually stuck.”

— Marguerite Doyle, Chief Operating Officer, Atlas Manufacturing Co.

Note: Project PRJ-2201 closed with a final budget variance; see Financial Summary FY2025 and the Atlas Manufacturing SOW Amendment for the audited actual cost figure, which exceeded the original signed budget by approximately 5.5% due to an extended Line 1 rollout window.